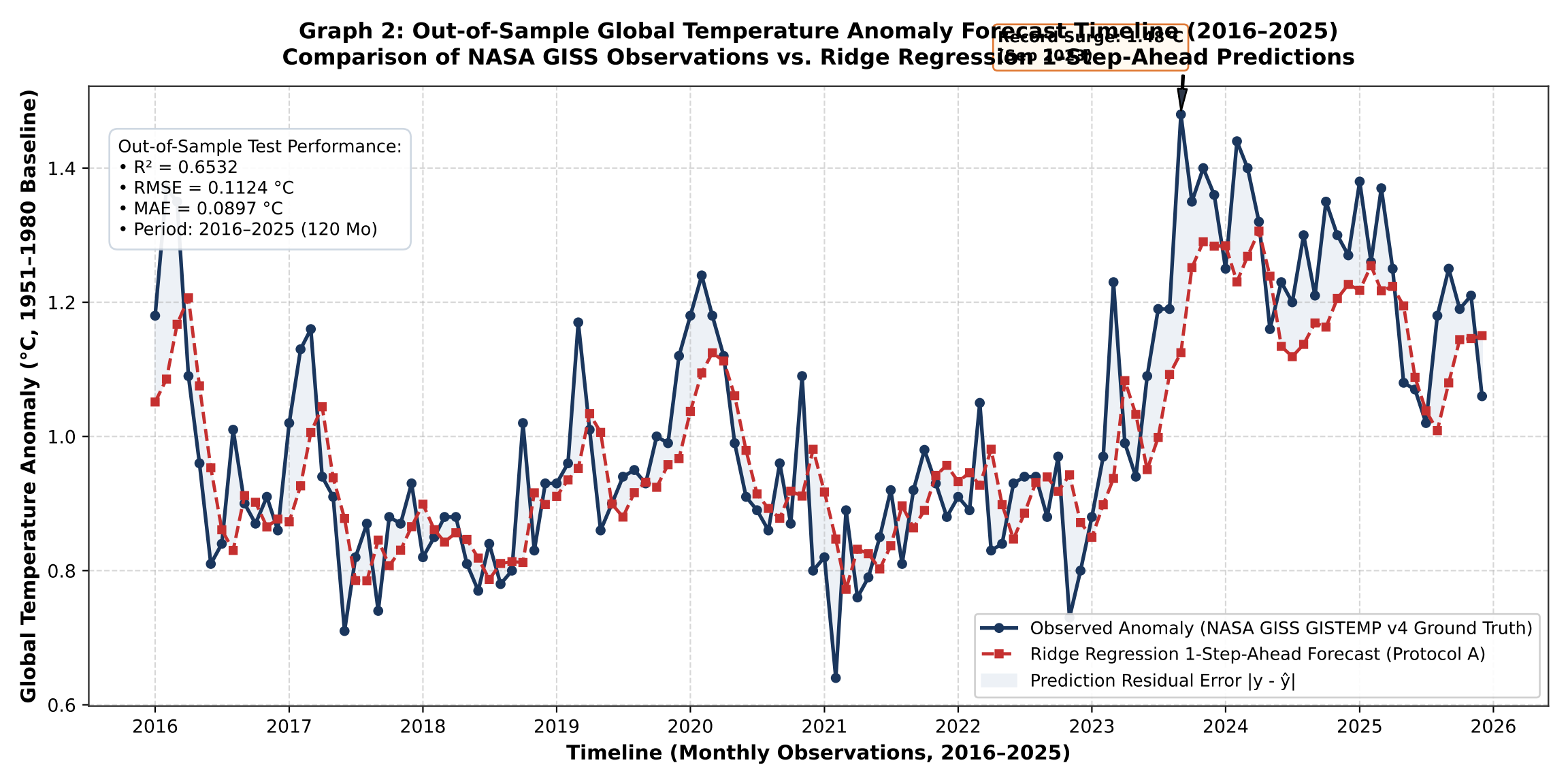
Graph 2: Out-of-Sample Global Temperature Anomaly Forecast Timeline (2016–2025)
Comparison of NASA GISS Observations vs. Ridge Regression 1-Step-Ahead Predictions
Record Surge: 1.48 °C
Jan 2024
Out-of-Sample Test Performance:
• R² = 0.6532
• RMSE = 0.1124 °C
• MAE = 0.0897 °C
• Period: 2016–2025 (120 Mo)
Observed Anomaly (NASA GISS GISTEMP v4 Ground Truth)
Ridge Regression 1-Step-Ahead Forecast (Protocol A)
Prediction Residual Error |y - ŷ|
Global Temperature Anomaly (°C, 1951–1980 Baseline)
0.6
0.8
1.0
1.2
1.4
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Timeline (Monthly Observations, 2016–2025)